
VIABILITY-GATED SKILL RESIDUALS FOR TEST-TIME LOCOMOTION CONTROL

A PREPRINT

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July 2026

ABSTRACT

Skill discovery can produce reusable temporal abstractions, but directly replaying open-loop action chunks is brittle for contact-rich legged locomotion. We introduce *Viability-Gated Skill Residuals* (VGSR), a test-time control framework that builds an online archive from a converged locomotion policy and reuses archived behaviors as command-space residuals over a pretrained low-level controller. At each high-level step, VGSR selects archive chunks using predicted goal progress and state applicability, converts the selected chunk into a velocity-command prior, and activates it only when a viability gate predicts that the residual is goal-aligned, stable, and energetically in-distribution. In Isaac Lab rough-terrain experiments, VGSR improves a strong pretrained Unitree Go2 controller on harder target-reaching tasks, reducing an energy proxy by 9.3% (1.99 vs. 2.20) while maintaining a small success-rate advantage (0.79 vs. 0.78) in a 20-trial-per-target aggregate of 100 episodes per method. On the H1 humanoid, VGSR preserves stability in a regime where the direct controller is already saturated. We further show that viability can be learned from decision-level features (test AUC 0.97 under an episode-group split), enabling a learned gate that remains stable on held-out target directions. Ablations show that removing the gate inflates energy consumption by over 50%, confirming that gating—not the archive alone—is the critical component. Our results suggest that skill archives are most useful for locomotion when used as guarded command priors rather than raw action replay.

Keywords skill discovery · legged locomotion · residual control · quality-diversity archives · reinforcement learning

1 Introduction

Deep reinforcement learning has produced remarkably capable low-level controllers for legged robots: massively parallel simulation makes it possible to train velocity-conditioned locomotion policies for quadrupeds and humanoids in minutes [Rudin et al., 2022, Makoviychuk et al., 2021], and such policies transfer to challenging real-world terrain [Lee et al., 2020, Miki et al., 2022, Hwangbo et al., 2019]. At the same time, a long line of work on temporal abstraction argues that agents should *reuse* behavior: options [Sutton et al., 1999, Bacon et al., 2017], unsupervised skill discovery [Eysenbach et al., 2019, Sharma et al., 2020], skill priors extracted from offline data [Pertsch et al., 2020, Ajay et al., 2021], and quality-diversity behavior repertoires [Mouret and Clune, 2015, Cully et al., 2015, Pugh et al., 2016] all build libraries of reusable behaviors.

These two threads meet awkwardly in contact-rich locomotion. The natural way to reuse a discovered skill—replaying its action sequence as a temporally extended option—is brittle when the low-level dynamics are sensitive to feedback. We observe two dominant failure modes. First, *initial-state mismatch*: when the state in which a skill was recorded differs from the current state, open-loop replay drives the robot into unrecoverable postures. Second, *open-loop error accumulation*: even from a well-matched initial state, joint-action sequences diverge quickly on rough terrain, under different headings, or under different commanded speeds. A third, more practical obstacle is that *the baseline is strong*: publicly available pretrained controllers, such as the RSL-RL policies distributed with Isaac Lab [Mittal et al., 2023,

Schwarke et al., 2025], already solve many target-reaching locomotion tasks, so any behavior-reuse mechanism must demonstrate value *on top of* a strong pretrained controller rather than against random or zero-action baselines.

This paper asks a correspondingly sharpened question:

Can an action archive extracted from a converged locomotion policy serve as a test-time behavioral memory that helps a strong pretrained controller on harder goals—without destabilizing it?

We answer with **Viability-Gated Skill Residuals (VGSR)**, a test-time control framework with three components. (i) *RL-converged online archive*: instead of populating a skill library from random exploration, we collect rollouts from a converged locomotion policy, segment them into fixed-horizon chunks, describe each chunk with a body-frame locomotion descriptor, and maintain an online archive filtered by utility and descriptor novelty—close in spirit to a quality-diversity repertoire [Cully et al., 2015]. (ii) *Command-space residuals*: at test time, archived chunks are never replayed as joint actions. The selected chunk’s velocity descriptor is converted into a high-level velocity command that is *blended* with the direct goal-directed command of a pretrained velocity-conditioned controller. The low-level policy retains full authority over posture, contacts, and joint control. (iii) *Viability gating*: the residual is activated only when it passes a viability gate checking goal alignment, predicted progress, chunk stability, an energy z-score, and a speed cap; otherwise the controller falls back to the direct command. We additionally train a lightweight viability classifier on logged decision features and show it can partially replace hand-tuned thresholds.

We evaluate on Isaac Lab rough-terrain target-reaching with the Unitree Go2 quadruped and the H1 humanoid, using the public pretrained RSL-RL checkpoints as both the archive source and the low-level controller, and—critically—as the baseline. Our claims are deliberately calibrated to the evidence. On harder Go2 targets, a 100-episode-per-method aggregate shows that VGSR reduces the energy proxy (1.99 vs. 2.20, bootstrap 95% CIs [1.88, 2.09] vs. [2.08, 2.31]) while preserving a marginal success advantage (0.79 vs. 0.78); final distance and final height are not uniformly better. On H1 the direct controller is saturated and VGSR neither helps nor destabilizes. Gate ablations show that the unguarded residual inflates energy by over 50%, and a learned gate is stable on held-out target directions. We also report negative results—most notably a persistent low-height failure on long forward targets—because they delineate what post-hoc command gating can and cannot fix.

Contributions. (1) An online skill/action archive built from a *converged* RL locomotion policy, with body-frame descriptors and utility-and-novelty filtering (Section 3.1). (2) A test-time control scheme that converts archived chunks into *command-space residuals* over a pretrained low-level controller, avoiding open-loop action replay (Section 3.3). (3) A *viability gate*—hand-designed and learned variants—that decides when the residual is safe and useful, together with an empirical study on Go2 and H1 rough-terrain tasks that honestly separates where VGSR helps (energy, marginal success on harder targets), where it is neutral (saturated H1), and where it fails (height-limited long targets) (Sections 3.4–5).

2 Related Work

Options and temporal abstraction. The options framework [Sutton et al., 1999] formalizes temporally extended actions, and option-critic [Bacon et al., 2017] learns option policies end-to-end; data-efficient hierarchical RL further layers goal-conditioned controllers [Nachum et al., 2018]. VGSR shares the motivation of temporal abstraction but does not learn option policies from scratch: it extracts reusable chunks from an already-converged locomotion policy and reuses them only at the command level.

Unsupervised skill discovery. DIAYN [Eysenbach et al., 2019] and DADS [Sharma et al., 2020] discover diverse skills via information-theoretic objectives without task reward. Our archive is instead built from task-driven, converged-policy rollouts; diversity enters only through descriptor-space clustering, and the archive is explicitly filtered for utility and stability rather than diversity per se.

Skill priors from offline data. SPiRL [Pertsch et al., 2020] and OPAL [Ajay et al., 2021] learn latent skill spaces and priors from offline experience to accelerate downstream RL; ASE learns large-scale reusable embeddings for simulated characters [Peng et al., 2022]. These methods train new high-level policies over the skill space. VGSR requires no downstream training: the archive is used directly at test time, and the “prior” takes the form of a gated command residual.

Quality-diversity repertoires. MAP-Elites [Mouret and Clune, 2015] and behavior repertoires enable rapid damage adaptation by searching a pre-built archive at deployment [Cully et al., 2015, Pugh et al., 2016]. VGSR borrows the descriptor-archive idea but differs in how the archive acts: rather than switching between archived controllers, archived chunks modulate the command input of a single pretrained policy.

Residual control. Residual policy learning adds a learned corrective action on top of a base controller [Silver et al., 2018, Johannink et al., 2019]. VGSR is residual in command space rather than action space, and the residual is not learned by gradient descent—it is *retrieved* from an archive and gated by viability checks.

Command-conditioned legged locomotion. Massively parallel training yields robust velocity-tracking policies [Rudin et al., 2022, Makoviychuk et al., 2021, Mittal et al., 2023, Schwarke et al., 2025], which transfer to harsh real terrain [Lee et al., 2020, Miki et al., 2022] and adapt online [Kumar et al., 2021]; multiplicity-of-behavior training exposes gait diversity through command interfaces [Margolis and Agrawal, 2022]. We treat such a policy as a fixed, strong low-level layer and study whether an archive distilled from its own converged behavior can improve it at test time. Recent imitation approaches also exploit action chunking [Zhao et al., 2023], but execute chunks closed-loop with dense sensing; our negative results (Section 5.6) reinforce that raw chunk replay is unreliable for rough-terrain locomotion.

The individual ingredients—archives, residuals, gating—are each established. The contribution here is the specific combination for contact-rich locomotion, evaluated directly against the strong pretrained controller it augments.

3 Method

VGSR has three stages: (i) building an online archive from a converged policy, (ii) selecting archive chunks at test time and converting them to command residuals, and (iii) gating residual activation by viability. Throughout, π_θ denotes a pretrained velocity-conditioned low-level policy that maps observations and a 3-D command $c = (v_x, v_y, \omega_z)$ (body-frame forward velocity, lateral velocity, yaw rate) to joint actions.

3.1 RL-Converged Online Skill Archive

Data collection. We roll out the converged policy π_θ under its training command distribution and record observations, actions, rewards, base poses, base velocities, joint velocities, and termination flags. Trajectories are segmented into fixed-horizon chunks τ_i with a fixed stride.

Body-frame locomotion descriptors. Each chunk is compressed into a descriptor d_i expressed in the body frame of its initial state: horizontal displacement $(\Delta x, \Delta y)$, yaw change $(\Delta \psi)$, average forward/lateral velocity and yaw rate, mean body height and height standard deviation, mean action norm and joint-velocity norm, an energy proxy

$$e_i = \text{mean}(|a_t \odot \dot{q}_t|), \quad (1)$$

a stability score combining posture, height, and termination statistics, and an action-difference smoothness score. Body-frame descriptors make “walk forward,” “strafe,” and “turn” invariant to world heading, so the same behavior clusters together across rollouts. Each chunk also stores an initial-state feature f_i and a scalar utility u_i aggregating reward, displacement, stability, smoothness, and energy.

Online archive update. The archive retains at most M chunks. Unstable chunks that are not descriptor-novel are filtered; the remainder are ranked by utility and the top- M retained. Chunks are clustered by weighted k -means in descriptor space, with weights emphasizing motion outcomes $(\Delta x, \Delta y, \Delta \psi)$, average velocities, yaw rate, stability, smoothness) and down-weighting energy and action norms so that near-identical motions are not split by energy noise. Central and high-utility representatives are kept per cluster. The result is a quality-diversity-style behavioral memory [Cully et al., 2015] whose entries are, by construction, behaviors the converged policy can actually execute stably. Algorithm 1 summarizes the procedure.

3.2 Archive Chunk Selection

At test time the robot must reach a target position g . At each high-level step, the controller computes the body-frame target g_{body} and scores every archive chunk k by

$$J_k = \hat{D}_k + \lambda_{\text{state}} \|f_{\text{now}} - f_k\| + \lambda_{\text{energy}} e_k + \lambda_{\text{stab}} (1 - s_k) + \lambda_{\text{np}} \max(0, -\hat{P}_k) - \lambda_{\text{util}} z(u_k) - \lambda_{\text{prog}} \max(0, \hat{P}_k), \quad (2)$$

where $\hat{D}_k$ is the predicted distance to the target after executing chunk k (obtained by applying the chunk’s body-frame displacement descriptor to the current pose), $\hat{P}_k$ is the predicted progress (current distance minus $\hat{D}_k$), f_k is the chunk’s initial-state feature, e_k its energy proxy, s_k its stability score, and $z(u_k)$ the archive-normalized utility. The state-distance term implements *applicability*: chunks recorded in dissimilar states are penalized, addressing the initial-state-mismatch failure mode directly in the selection objective. Descriptor-based prediction plays the role of a one-step motion model without learning full dynamics.

Algorithm 1 Build Online Action Archive from a Converged RL Policy**Require:** converged policy π_θ , environment E , horizon H , archive size M , clusters K

- 1: Collect rollouts of π_θ in E ; record observations, actions, rewards, poses, velocities, terminations
- 2: Segment trajectories into fixed-horizon chunks $\{\tau_i\}$ with stride
- 3: **for** each chunk τ_i **do**
- 4: $d_i \leftarrow$ body-frame locomotion descriptor of τ_i ; $f_i \leftarrow$ initial-state feature
- 5: $u_i \leftarrow$ utility(reward, displacement, stability, smoothness, energy)
- 6: **end for**
- 7: Filter unstable, non-novel chunks; keep top- M by utility
- 8: Cluster descriptors with weighted k -means (K clusters, outcome-weighted dimensions)
- 9: Keep central and high-utility representatives per cluster

Ensure: archive $\mathcal{A} = \{(\tau_i, d_i, f_i, u_i, \text{skill}_i)\}$ **3.3 Command-Space Residual Control**

The selected chunk $k^* = \arg \min_k J_k$ is *not* replayed. Instead, its velocity descriptor is read out as a candidate command

$$c_{\text{skill}} = (\bar{v}_x^{(k^*)}, \bar{v}_y^{(k^*)}, \bar{\omega}_z^{(k^*)}), \quad (3)$$

while the direct goal command is computed from the body-frame target as

$$c_{\text{goal}} = (\text{clip}(k_v x_{\text{local}}), \text{clip}(k_v y_{\text{local}}), \text{clip}(k_\psi \text{atan2}(y_{\text{local}}, x_{\text{local}}))). \quad (4)$$

The executed command blends the two:

$$c = (1 - \alpha) c_{\text{goal}} + \alpha c_{\text{skill}}, \quad \alpha = \begin{cases} \alpha_{\max} \text{clip}(\|g_{\text{body}}\|/d_{\text{blend}}, 0, 1) & \text{if viable,} \\ 0 & \text{otherwise,} \end{cases} \quad (5)$$

and π_θ executes with command c for a short horizon before the next high-level step. The distance-proportional blend automatically anneals the residual near the goal. Because π_θ retains full control of posture, contacts, and joints, the archive can only bias *where the policy is told to go*, never *how it moves its joints*—this is the central design decision that separates VGSR from open-loop skill replay.

3.4 Viability Gating

Heuristic gate. The residual is activated ($\alpha > 0$ in Eq. 5) only if the candidate passes all of: (i) *alignment*: the skill command direction agrees with the goal direction, $\text{align}(c_{\text{skill}}, g_{\text{body}}) \geq \epsilon_a$; (ii) *progress*: predicted progress $\hat{P}_{k^*} \geq \epsilon_p$; (iii) *stability*: chunk stability $s_{k^*} \geq \epsilon_s$; (iv) *energy*: the chunk’s energy z-score within the archive does not exceed ϵ_e ; (v) *speed cap*: $\|c_{\text{skill}}\| \leq (1 + \rho)\|c_{\text{goal}}\|$. A recovery clause keeps the controller active near the goal when body height is below a threshold fraction of nominal, prioritizing height recovery over premature stopping. On failure of any check, the controller falls back to the pure direct command—the pretrained policy is the safe default.

Learned viability discriminator. To reduce threshold hand-tuning, we log every candidate decision during guarded evaluation and train a lightweight classifier p_ϕ mapping decision features to episode-level success. Features include current/predicted distance, predicted progress, state distance and its RBF applicability, utility, energy, stability, alignment, energy z-score, and the norms of both commands. To avoid merely cloning the heuristic, the deployable variant *excludes* the heuristic gate’s own outputs (its blend weight, accept flag, and low-height flag). Train/test splitting is by *episode group*, so decisions from one episode never appear on both sides. At deployment the learned gate can *replace* the heuristic, be *union*-ed with it, or *intersect* it; intersection—the learned model further filtering candidates that already pass the physical guard—is the most stable and is what we report. Algorithm 2 summarizes the test-time loop.

4 Experimental Setup

Environments and robots. All experiments use Isaac Lab [Mittal et al., 2023] rough-terrain velocity environments: Isaac-Velocity-Rough-Unitree-Go2-v0 (quadruped) and Isaac-Velocity-Rough-H1-v0 (humanoid). The low-level controller is the publicly distributed pretrained RSL-RL checkpoint for each task [Rudin et al., 2022, Schwarke et al., 2025]; the same checkpoint generates the archive rollouts. The task is target reaching: the robot must move its base to a target position g specified relative to its start pose.

Algorithm 2 Viability-Gated Skill Residual Control**Require:** pretrained policy π_θ , archive $\mathcal{A}$, target g , optional learned gate p_ϕ

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1: while episode not done do
2:   Observe state; compute body-frame target  $g_{\text{body}}$  and direct command  $c_{\text{goal}}$ 
3:   for each chunk  $k \in \mathcal{A}$  do
4:     Predict displacement from  $d_k$ ; compute  $\hat{D}_k, \hat{P}_k$ , state applicability, and cost  $J_k$  (Eq. 2)
5:   end for
6:    $k^* \leftarrow \arg \min_k J_k$ ;  $c_{\text{skill}} \leftarrow$  velocity descriptor of  $k^*$ 
7:    $h \leftarrow$  heuristic gate( $k^*$ ): alignment  $\wedge$  progress  $\wedge$  stability  $\wedge$  energy  $\wedge$  speed cap
8:   if learned gate enabled then
9:      $h \leftarrow h \wedge [p_\phi(\text{decision features}) \geq \eta]$   $\triangleright$  intersect mode
10:  end if
11:   $\alpha \leftarrow h ? \alpha_{\text{max}} \text{clip}(\|g_{\text{body}}\|/d_{\text{blend}}, 0, 1) : 0$ 
12:  Execute  $\pi_\theta$  with  $c = (1 - \alpha)c_{\text{goal}} + \alpha c_{\text{skill}}$  for a short horizon
13: end while

```

Table 1: Go2 rough terrain, moderate targets (10 trials $\times$ 4 targets). Arrows give the preferred direction.

Method	Success $\uparrow$	Final dist. (m) $\downarrow$	Energy $\downarrow$	Final height (m)	Term. $\downarrow$
VGSR (ours)	1.000	0.2892	1.8924	0.3588	0.000
direct	0.975	0.2893	1.8147	0.3604	0.000

Target sets. *Moderate* Go2 targets: (0.75, 0), (0.75, 0.75), (1.5, 0), (0, 0.75) m. *Harder* Go2 targets: (2, 0), (1.5, 1), (0, 1.25), (−0.75, 0), (1, −1) m, including backward and lateral goals that the velocity-tracking training distribution covers less densely. *Holdout* directions for learned-gate evaluation (never used for tuning): (1.25, 0.5), (0.5, 1), (−0.5, 0.5), (1.25, −0.5) m. H1 targets: (0.5, 0), (0.5, 0.5), (1, 0) m.

Methods. (1) *direct*: the pretrained controller driven by the direct target command only—a strong baseline, not a strawman; (2) *skill*: unguarded residual, i.e. Eq. 5 with the gate forced open; (3) *VGSR*: heuristic-gated residual (with recovery clause on harder targets); (4) *learned-VGSR*: heuristic gate intersected with the learned discriminator at threshold $\eta = 0.999$. We deliberately exclude random/zero-action baselines from the main comparison; they are sanity checks only, and beating them would say nothing given the strength of the pretrained controller.

Metrics. Success rate (no early termination, final height above threshold, final XY distance below threshold); average final distance (m); average energy proxy (Eq. 1); average final height (m); early-termination rate; and the number of high-level steps in which the residual was active. Note the success criterion includes a *final-height* requirement: an episode that reaches the target XY neighborhood in a crouched posture fails. For the strengthened harder-target aggregate we report episode-level bootstrap 95% confidence intervals.

5 Results

5.1 Go2 Rough Terrain: Moderate Targets

Table 1 reports 10 trials per target (40 episodes per method). VGSR attains a slightly higher success rate (1.000 vs. 0.975) at equal final distance, with a small energy overhead. The takeaway is modest by design: on targets the pretrained controller already handles well, the gated residual does no harm—the gate correctly stays out of the way.

5.2 Go2 Rough Terrain: Harder Targets (Main Result)

The initial 5-trial evaluation (25 episodes per method) that motivated the gate design showed VGSR improving success (0.800 vs. 0.760), final distance (0.2822 vs. 0.2884), and energy (2.09 vs. 2.39). Because 25 episodes is too few for a headline claim, we strengthened the evaluation to 20 trials per target (100 episodes per method) by aggregating three seeds. Table 2 reports means with episode-level bootstrap 95% CIs.

Three observations, stated at the strength the data supports. *First*, the energy improvement is the most consistent effect: 1.99 vs. 2.20 (−9.3%), with CIs that barely overlap. The archive supplies locally better-matched motion priors—for example, committed lateral or turning commands where the proportional direct command produces inefficient in-place correction. *Second*, the success advantage is marginal (0.790 vs. 0.780, heavily overlapping CIs): the initial 5-trial

Table 2: Go2 rough terrain, harder targets, strengthened aggregate (20 trials $\times$ 5 targets = 100 episodes per method; 3 seeds). Brackets are episode-level bootstrap 95% CIs.

Method	Success $\uparrow$	Final dist. (m) $\downarrow$	Energy $\downarrow$	Final height (m)
VGSR (ours)	0.790 [0.710, 0.870]	0.2979 [0.2749, 0.3370]	1.9913 [1.8819, 2.0936]	0.3071 [0.2885, 0.3237]
direct	0.780 [0.700, 0.860]	0.2888 [0.2874, 0.2902]	2.1954 [2.0831, 2.3110]	0.3174 [0.3027, 0.3313]

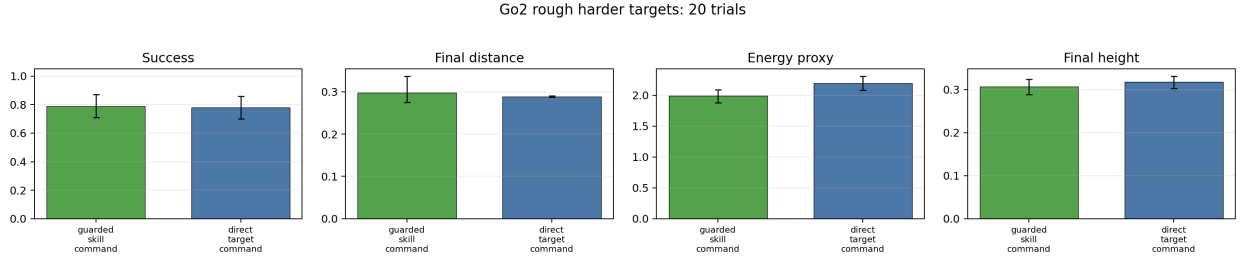


Figure 1: Go2 harder targets, strengthened 20-trial aggregate (100 episodes per method). VGSR reduces the energy proxy with overlapping-but-shifted CIs while success is marginally higher; final distance and final height are slightly worse, driven by the height-limited 2 m forward target (Section 5.6).

gap of +0.04 did not survive aggregation, and we report this honestly rather than leaning on the favorable seed. *Third*, VGSR’s final distance and height are slightly worse on average; both effects trace mostly to the 2 m forward target where episodes approach the XY goal but fail the height criterion (Section 5.6). Figure 1 visualizes the aggregate; Figure 2 shows representative trajectories.

5.3 Gate Ablation: Is the Gate Necessary?

Table 3 isolates the gate’s contribution on harder targets (5 trials, seed 102). The unguarded residual (*skill*) gets closest to the target (0.2739 m) but at an energy of 3.18—more than 50% above either gated variant—because it fires the residual in 18.6 high-level steps per episode on average, nearly twice the gated rate (10.3). The heuristic gate restores energy parity with the direct controller (2.08 vs. 2.10) while keeping most of the distance benefit. On this seed the direct controller has the best success rate (0.92), and VGSR suffered one early termination on the 2 m target: the gate is *necessary* to suppress high-energy overuse of the archive, but the current heuristic form is *not sufficient* for a uniform win—consistent with the aggregate picture in Table 2.

5.4 Learned Viability Gate

Can viability be learned? Training on 2,879 logged decisions across 90 episode groups with an episode-group split (1,900 train / 979 test decisions; 68/22 groups), the deployable classifier—with all heuristic-gate outputs excluded from its inputs—reaches test accuracy 0.913 and test AUC 0.973. Decision-level features therefore carry substantial viability signal beyond the hand-designed thresholds.

Held-out directions. On four holdout targets (Table 4), all methods are saturated at success 1.0; learned-VGSR (intersect, $\eta = 0.999$) reduces energy relative to heuristic VGSR (1.9096 vs. 1.9346) and raises final height, while remaining statistically indistinguishable from the direct controller. The learned gate is *stable* on directions it never saw—the key deployment property—though not yet clearly better than direct control here.

Harder targets expose the label problem. On harder targets (Table 5, seed 103), learned-VGSR achieves the best final distance (0.2779) and energy (1.9232) but the *worst* success (0.760), losing episodes on the 2 m forward target through the height criterion. The diagnosis is instructive: episode-level success labels teach the classifier that “low-energy, goal-approaching” decisions are good, which correlates with—but is not identical to—“arriving at a safe height.” The learned gate optimizes the correlation and misses the causal requirement. Mode ablations reinforce the point: replace mode at threshold 0.5 over-accepts residuals and underperforms direct control; even at 0.999 it remains too aggressive. Intersection with the physical guard is currently the only reliable deployment mode.

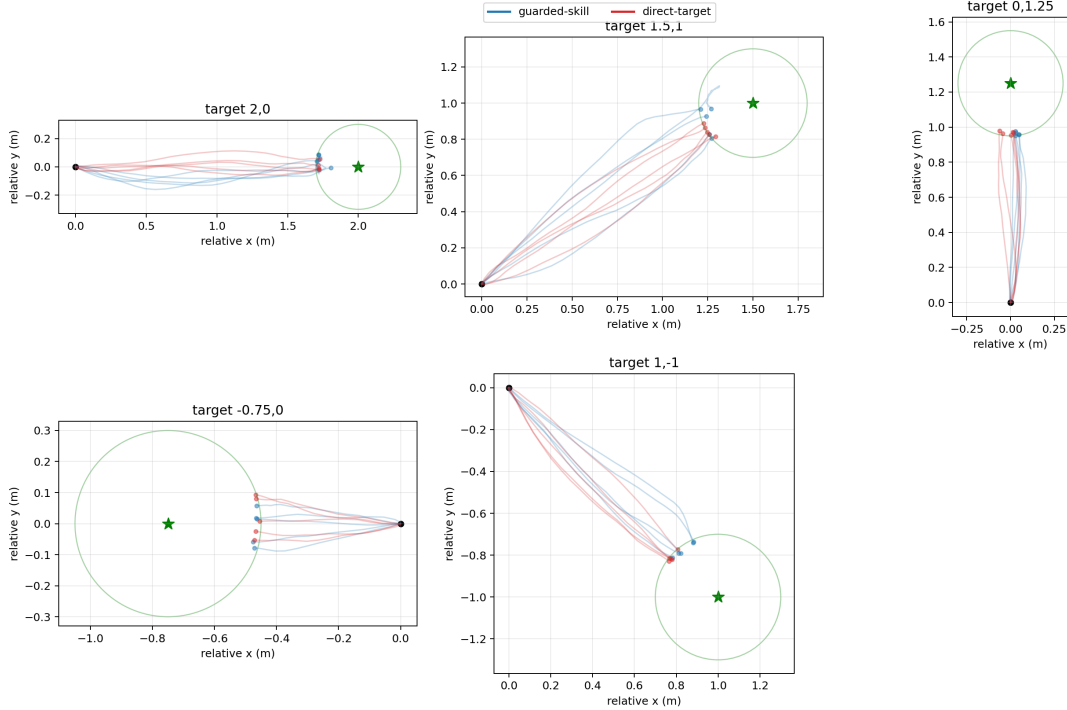


Figure 2: Go2 harder-target trajectories (blue: VGSR; red: direct). Both controllers reach the target neighborhood (green circle) on most targets; qualitative differences are largest on the diagonal (1.5, 1) and lateral (0, 1.25) goals where the residual commits to smoother approach paths.

Table 3: Gate ablation on Go2 harder targets (5 trials $\times$ 5 targets, seed 102). “Resid. used” is the average number of high-level steps with an active residual.

Method	Success $\uparrow$	Final dist. (m) $\downarrow$	Energy $\downarrow$	Final height (m)	Term. $\downarrow$	Resid. used
skill (no gate)	0.800	0.2739	3.1806	0.3205	0.000	18.64
VGSR (ours)	0.840	0.3570	2.0802	0.3278	0.040	10.28
direct	0.920	0.2889	2.0988	0.3372	0.000	0.00

5.5 Cross-Embodiment: H1 Humanoid

On H1 rough-terrain targets (Table 6), the direct controller is already saturated (success 1.0), leaving no headroom. VGSR matches success and final distance (marginally better, 0.2905 vs. 0.2911), stands slightly taller, and costs slightly more energy. We present this strictly as a *stability-of-transfer* result: the same archive pipeline, descriptors, and gate applied to a humanoid neither destabilize the controller nor degrade its performance. It is not evidence of improvement, and we do not claim any.

5.6 Negative Results and Failure Modes

We report five negative results that shaped the design and bound the claims.

(1) Raw action replay fails. Replaying archived joint-action chunks open-loop—the classic “skill as option” execution—reliably destabilizes both Go2 and H1 on rough terrain, through the initial-state-mismatch and error-accumulation modes of Section 1. This failure is why VGSR operates in command space at all.

(2) The 2 m forward target is height-limited. On long forward targets the robot frequently reaches the XY neighborhood but finishes below the height threshold and fails. Reactive low-height gates were tested at two settings: an aggressive variant preserves height but suppresses useful progress; a balanced variant preserves energy but still fails the height criterion. Post-hoc command gating is too late once the base policy has entered a crouched, low-viability regime—recovery must be predicted *before* it is needed, not reacted to.

Table 4: Learned-gate evaluation on holdout target directions (5 trials $\times$ 4 targets).

Method	Success $\uparrow$	Final dist. (m) $\downarrow$	Energy $\downarrow$	Final height (m)	Resid. used
learned-VGSR (intersect)	1.000	0.2911	1.9096	0.3622	8.30
VGSR (heuristic)	1.000	0.2877	1.9346	0.3577	10.50
direct	1.000	0.2890	1.9063	0.3585	0.00

Table 5: Learned gate on Go2 harder targets (5 trials $\times$ 5 targets, seed 103).

Method	Success $\uparrow$	Final dist. (m) $\downarrow$	Energy $\downarrow$	Final height (m)	Resid. used
learned-VGSR (intersect)	0.760	0.2779	1.9232	0.3093	7.76
VGSR (heuristic)	0.800	0.2865	2.0267	0.3180	8.52
direct	0.800	0.2896	2.0812	0.3333	0.00

(3) **Learned gate in replace mode over-accepts** (Section 5.4), at both $\eta = 0.5$ and $\eta = 0.999$.

(4) **Weaker blending is not automatically safer.** Lowering the residual blend cap to 0.25 caused an unstable right-diagonal holdout target in one seed; residual strength and gating interact nonlinearly, so “smaller residual” is not a substitute for a better gate.

(5) **The archive alone is not the contribution.** Table 3 shows that an ungated archive is a net negative on energy. Any claim that a skill archive helps locomotion must be paired with an activation mechanism; in our experiments the gate, not the archive, is the binding constraint.

6 Limitations and Future Work

Scope of the empirical claim. The positive result is an energy reduction with a marginal success advantage on harder Go2 targets against a strong pretrained controller, plus gate-ablation and holdout-stability evidence. It is not an across-the-board win: final distance and height are not uniformly better, H1 shows no improvement, and one gate-ablation seed favors the direct controller on success. Evaluations use 25–100 episodes per method per setting; wider multi-seed sweeps would tighten the intervals.

Episode-level viability labels are too coarse. The learned gate’s failure on the 2 m target traces to labels that reward correlates of success rather than height-safe arrival. The natural fix is short-horizon labels: for each logged decision, compare executing the residual versus the direct command over the next N steps on progress, height safety, and energy-normalized progress, turning the discriminator into a genuine applicability model.

Post-hoc gating cannot rescue an unviable base state. Once the low-level policy is crouched, no command filter recovers it (negative result 2). Complementary directions: a viability model that predicts height collapse before residual execution, or a base policy trained with stronger upright-recovery objectives.

Breadth. All results are in simulation, on target reaching, with one archive configuration per robot. Archive-size ablations, path following, disturbance recovery, stairs/slopes, and hardware transfer are open. The framework is low-level-controller-agnostic in principle—any velocity-conditioned policy exposes the required interface [Margolis and Agrawal, 2022, Kumar et al., 2021]—but we have only instantiated RSL-RL checkpoints.

7 Conclusion

We presented VGSR, a test-time framework that turns rollouts of a converged locomotion policy into an online behavioral archive and reuses that archive as viability-gated, command-space residuals over the same pretrained controller. Across Isaac Lab rough-terrain experiments the evidence supports a calibrated conclusion: raw action replay is unreliable; command-space residuals are the right interface; viability gating is necessary to prevent energetic overuse of the archive; and the combination yields consistent energy savings with marginal success gains on harder Go2 targets while remaining neutral on a saturated H1 task. Viability itself is learnable from decision features, though episode-level labels are not yet sufficient for height-safe deployment. The broader suggestion is that for contact-rich locomotion, skill archives are best understood not as libraries of replayable behaviors but as *guarded command priors* over strong low-level controllers.

Table 6: H1 humanoid, rough terrain (5 trials $\times$ 3 targets). The direct controller is saturated; the result demonstrates transfer stability, not improvement.

Method	Success	Final dist. (m) $\downarrow$	Energy $\downarrow$	Final height (m)	Term. $\downarrow$
VGSR (ours)	1.000	0.2905	0.5394	1.0169	0.000
direct	1.000	0.2911	0.4971	1.0091	0.000

Reproducibility

All experiments use public Isaac Lab environments and public pretrained RSL-RL checkpoints. Archive construction, evaluation, viability-model training, and plotting code, together with all result files aggregated in this paper, are available in the project repository.

References

- Nikita Rudin, David Hoeller, Philipp Reist, and Marco Hutter. Learning to walk in minutes using massively parallel deep reinforcement learning. In *Conference on Robot Learning (CoRL)*, pages 91–100, 2022. arXiv:2109.11978.
- Viktor Makoviychuk, Lukasz Wawrzyniak, Yunrong Guo, Michelle Lu, Kier Storey, Miles Macklin, David Hoeller, Nikita Rudin, Arthur Allshire, Ankur Handa, and Gavriel State. Isaac Gym: High performance GPU based physics simulation for robot learning. In *NeurIPS Datasets and Benchmarks Track*, 2021. arXiv:2108.10470.
- Joonho Lee, Jemin Hwangbo, Lorenz Wellhausen, Vladlen Koltun, and Marco Hutter. Learning quadrupedal locomotion over challenging terrain. *Science Robotics*, 5(47):eabc5986, 2020.
- Takahiro Miki, Joonho Lee, Jemin Hwangbo, Lorenz Wellhausen, Vladlen Koltun, and Marco Hutter. Learning robust perceptive locomotion for quadrupedal robots in the wild. *Science Robotics*, 7(62):eabk2822, 2022.
- Jemin Hwangbo, Joonho Lee, Alexey Dosovitskiy, Dario Bellicoso, Vassilios Tsounis, Vladlen Koltun, and Marco Hutter. Learning agile and dynamic motor skills for legged robots. *Science Robotics*, 4(26):eaau5872, 2019.
- Richard S. Sutton, Doina Precup, and Satinder Singh. Between MDPs and semi-MDPs: A framework for temporal abstraction in reinforcement learning. *Artificial Intelligence*, 112(1-2):181–211, 1999.
- Pierre-Luc Bacon, Jean Harb, and Doina Precup. The option-critic architecture. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 31, 2017.
- Benjamin Eysenbach, Abhishek Gupta, Julian Ibarz, and Sergey Levine. Diversity is all you need: Learning skills without a reward function. In *International Conference on Learning Representations (ICLR)*, 2019. arXiv:1802.06070.
- Archit Sharma, Shixiang Gu, Sergey Levine, Vikash Kumar, and Karol Hausman. Dynamics-aware unsupervised discovery of skills. In *International Conference on Learning Representations (ICLR)*, 2020. arXiv:1907.01657.
- Karl Pertsch, Youngwoon Lee, and Joseph J. Lim. Accelerating reinforcement learning with learned skill priors. In *Conference on Robot Learning (CoRL)*, 2020. arXiv:2010.11944.
- Anurag Ajay, Aviral Kumar, Pulkit Agrawal, Sergey Levine, and Ofir Nachum. OPAL: Offline primitive discovery for accelerating offline reinforcement learning. In *International Conference on Learning Representations (ICLR)*, 2021. arXiv:2010.13611.
- Jean-Baptiste Mouret and Jeff Clune. Illuminating search spaces by mapping elites. *arXiv preprint arXiv:1504.04909*, 2015.
- Antoine Cully, Jeff Clune, Danesh Tarapore, and Jean-Baptiste Mouret. Robots that can adapt like animals. *Nature*, 521(7553):503–507, 2015.
- Justin K. Pugh, Lisa B. Soros, and Kenneth O. Stanley. Quality diversity: A new frontier for evolutionary computation. *Frontiers in Robotics and AI*, 3:40, 2016.
- Mayank Mittal, Calvin Yu, Qinxu Yu, Jingzhou Liu, Nikita Rudin, David Hoeller, Jia Lin Yuan, Ritvik Singh, Yunrong Guo, Hammad Mazhar, Ajay Mandlekar, Buck Babich, Gavriel State, Marco Hutter, and Animesh Garg. Orbit: A unified simulation framework for interactive robot learning environments. *IEEE Robotics and Automation Letters*, 8(6):3740–3747, 2023. arXiv:2301.04195.
- Clemens Schwarke, Mayank Mittal, Nikita Rudin, David Hoeller, and Marco Hutter. RSL-RL: A learning library for robotics research. *arXiv preprint arXiv:2509.10771*, 2025.
- Ofir Nachum, Shixiang Gu, Honglak Lee, and Sergey Levine. Data-efficient hierarchical reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 31, 2018.

- Xue Bin Peng, Yunrong Guo, Lina Halper, Sergey Levine, and Sanja Fidler. ASE: Large-scale reusable adversarial skill embeddings for physically simulated characters. In *ACM Transactions on Graphics (SIGGRAPH)*, volume 41, 2022. arXiv:2205.01906.
- Tom Silver, Kelsey Allen, Josh Tenenbaum, and Leslie Kaelbling. Residual policy learning. *arXiv preprint arXiv:1812.06298*, 2018.
- Tobias Johannink, Shikhar Bahl, Ashvin Nair, Jianlan Luo, Avinash Kumar, Matthias Loskyll, Juan Aparicio Ojea, Eugen Solowjow, and Sergey Levine. Residual reinforcement learning for robot control. In *IEEE International Conference on Robotics and Automation (ICRA)*, pages 6023–6029, 2019.
- Ashish Kumar, Zipeng Fu, Deepak Pathak, and Jitendra Malik. RMA: Rapid motor adaptation for legged robots. In *Robotics: Science and Systems (RSS)*, 2021. arXiv:2107.04034.
- Gabriel B. Margolis and Pulkrit Agrawal. Walk these ways: Tuning robot control for generalization with multiplicity of behavior. In *Conference on Robot Learning (CoRL)*, 2022. arXiv:2212.03238.
- Tony Z. Zhao, Vikash Kumar, Sergey Levine, and Chelsea Finn. Learning fine-grained bimanual manipulation with low-cost hardware. In *Robotics: Science and Systems (RSS)*, 2023. arXiv:2304.13705.